

Ibrahim Hossain Sajal

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EDUCATION

PhD in Data Science & Statistics

School of Natural Science and Mathematics, University of Texas at Dallas

Aug. 2019 – Dec. 2024

Masters in Applied Statistics

Institute of Statistical Research and Training, University of Dhaka, Bangladesh

Jan. 2017 – Dec. 2017

Bachelors in Applied Statistics

Institute of Statistical Research and Training, University of Dhaka, Bangladesh

Jan. 2013 – Dec. 2016

PROFESSIONAL EXPERIENCE

Postdoctoral Fellow

Jan. 2025 – Present

Division of Cancer Epidemiology & Genetics, National Cancer Institute, National Institutes of Health

- Investigated proteomic intermediaries linking risk factors to renal cancer using two-stage Mendelian Randomization (MR), complemented by mediation analysis, cis-MR, colocalization, and differential gene expression.
- Developed and implemented pipelines for a summary-statistics based multivariate test by aggregating multiple tests with complementary strength in capturing genetic association.
- Conducted multivariate analyses to identify associations with pan-cancer phenotypes, protein modules (WGCNA), and predefined pathways from KEGG and Hallmark.
- Identified pan-cancer associations with protein modules using Bayesian Non-Negative Matrix Factorization and regularization.
- Investigating the obesity paradox by examining how genetic predisposition to BMI influences RCC survival outcomes.

Graduate Research Assistant

Aug. 2019 – Dec. 2024

School of Natural Science and Mathematics, University of Texas at Dallas

- **CBCRisk-Black - Contralateral breast cancer (CBC) risk prediction model for Black women.**
 - * Developed and validated CBCRisk-Black, integrating matched LASSO logistic regression on BCSC data with SEER-based survival modeling to generate 3- and 5-year absolute risk estimates for clinical decision support.
 - * Improved predictive performance by 10% AUC over the existing CBCRisk model using six-fold cross-validation, demonstrating enhanced discrimination and robustness in an underrepresented patient population.
 - * Productionized the model via multiple imputation (MICE) for missing data resilience and deployed a user-facing R package to enable scalable implementation by clinicians, researchers, and health systems.
- **CBCRisk-Mastectomy - Pre-surgical CBC risk prediction tool to aid for informed surgery decision making.**
 - * Developed a pre-surgical CBC risk prediction tool that uses routinely available clinical factors at the time of mastectomy to generate individualized absolute risk estimates, enabling data-driven decision support for CPM.
- **Genetic association test - bivariate QBL: Detecting common and rare genetic association with two correlated phenotypes.**
 - * Applied Bayesian LASSO to regularize regression coefficients and latent variable to model correlation, improving signal detection through joint phenotype modeling.
 - * Implemented MCMC-based posterior inference and Bayes factor testing, achieving superior power over existing methods (e.g., Haplo.score) in simulation benchmarks.
 - * Validated performance using large-scale simulations and GAW19 exome sequencing data, identifying rare haplotypes associated with systolic and diastolic blood pressure.
- **Genetic association test - Multivariate QBL:**
 - * Developing a multivariate version of the bivariate QBL test using an efficient machine learning method, Variational Inference.
 - * Investigating the statistical properties of the test using simulations and comparing its power and efficiency to that of bivariate QBL.

Sr. Statistician

Jan. 2018 – Jun. 2019

James P Grant School of Public Health, BRAC University, Dhaka, Bangladesh

PRESENTATIONS

- **Meta-trans: A summary statistics-based test to identify trans-regulation of gene-sets by disease related variants.**
STATGEN 2026: Conference on Statistics in Genomics and Genetics, Atlanta, GA – (Accepted as Poster) May 2026
- **Integrative analysis identifies proteomic intermediates associated with renal cell carcinoma and its risk factors.**
 - *American Association for Cancer Research (AACR)*, San Diego, CA – (Accepted as Poster) Apr. 2026
 - *National Institutes of Health Research Fair*, Bethesda, MD Sep. 2025
- **Multivariate QBL for detecting rare haplotype association with correlated phenotypes.**
Joint Statistical Meetings (JSM), Nashville, TN Aug. 2025.
- **Bivariate QBL for Detecting Rare Haplotype Association with Two Correlated Phenotypes.**
 - *Advances in Statistical and Computational Methods for Analysis of Biomedical, Genetic, and Omics Data (ABGOD)*, Dallas, TX Mar. 2023
 - *Southern Regional Council on Statistics (SRCOS)*, Jekyll Island, GA Sep. 2022

TECHNICAL SKILLS

Data Analysis: R, Python, SAS, STATA, MATLAB

Programming Language: C++

Data management: SQL, Linux commands, HPC environment, Bash scripts, AWS

Documentation: Latex, Microsoft Word, Excel, PowerPoint, Power BI

Reference management: Endnote, Mendeley

AWARDS

- **Fellows Award for Research Excellence** – *National Institutes of Health* Jul. 2025
- **Mei Lein Fellowship** – *University of Texas at Dallas* May 2023, May 2022
- **Travel Award** – *ABGOD Conference, Dallas, TX*
- **Boyd Harshburger Travel Award** – *Southern Regional Conference on Statistics, Jekyll Island, GA* Sep. 2022
- **PhD Research Small Grant Program** – *University of Texas at Dallas* May 2022
- **Summer Institute in Statistical Genetics Scholarship** – *University of Washington* May 2021
- **Academic Excellence Award** – *University of Dhaka, Bangladesh* Dec. 2017

TRAINING AND WORKSHOPS

- **Python for Data Science Workshop** *University of Texas at Dallas* Jul. 2022
- **Summer Institute in Statistical Genetics** *University of Washington* Jul. 2021
Module: Introduction to Genetics and Genomics, Genetic Epidemiology, MCMC for Genetics.

VOLUNTEER EXPERIENCES

- **Advances in Statistical and Computational Methods for Analysis of Biomedical, Genetic, and Omics Data.** *Dallas, TX* Mar. 2023
- **Florence Nightingale Day** (Event to promote statistics to school-going students.) *Dallas, TX* Oct. 2022, Oct. 2021
- **Global Conference on Implementation Science (GCIS)** *Dhaka, Bangladesh* Jun. 2019
- **International Conference on Applied Statistics (ICAS)** *Dhaka, Bangladesh* Dec. 2014

PUBLICATIONS

1. **IH. Sajal**, A. Song, K. Brown, M. Machiella, P. Kraft, S. Chanock, M. Purdue, and D. Dutta (2026) *Integrative analysis prioritizes proteins associated to renal cell carcinoma and its risk factors*, **currently under review**.
2. **IH. Sajal**, R. Pfeiffer, I. Jatoi, M. Gail, R. Cecchini, P. Choudhary, and S. Biswas, (2026) *CBCRisk-Mastectomy: A Risk Prediction Tool to Aid for Contralateral Prophylactic Mastectomy Decision Making*, **currently under review**.
3. **IH. Sajal** and S. Biswas, (2023), *Bivariate Quantitative Bayesian LASSO for Detecting Association of Rare Haplotypes with Two Correlated Continuous Phenotypes*, *Frontiers in Genetics*, 14.
4. **IH. Sajal**, M. Chowdhury, T. Wang, D. Euhus, P. Choudhary, and S. Biswas, (2022), *CBCRisk-Black: A Personalized Contralateral Breast Cancer Risk Prediction Model for Black Women*, *Breast Cancer Research and Treatment*, 194(1):179-86.
5. R. Das Gupta, M. Akonde, **IH. Sajal**, A. Kibria, G. Muhammed, (2021), *Association between height and hypertension among US adults: analyses of National Health and Nutrition Examination Survey 2007–18*, *Clinical Hypertension*, 27(1), 1-12.
6. A. Talukder, R. Das Gupta, MR. Hashan, SS. Haider, **IH. Sajal**, M. Sarker, (2021), *Association between television viewing and overweight and obesity among women of reproductive age in Timor- Leste: evidence from the demographic health survey 2016*, *BMJ Open*, 11(8), e045547.
7. R. Das Gupta, M. Jahan, M. Hasan, I. Sutradhar, **IH. Sajal**, SS. Haider, M. Sarker, (2020), *Factors associated with tobacco use among Nepalese men aged 15–49 years: data from Nepal demographic and Health Survey 2016*, *Clinical Epidemiology and Global Health*, 8(3), 748-757.
8. R. Das Gupta, SS. Haider, I. Sutradhar, MR. Hashan, **IH. Sajal**, M. Sarker, (2019), *Association of frequency of television watching with overweight and obesity among women of reproductive age in India: evidence from a nationally representative study*, *PloS one*, 14(8), e0221758.
9. R. Das Gupta, SS. Haider, MR. Hashan, M. Hasan, I. Sutradhar, **IH. Sajal**, H. Joshi, MR. Haider, M. Sarker, (2019), *Association between the frequency of television watching and overweight and obesity among women of reproductive age in Nepal: analysis of data from Nepal Demographic Health survey 2016*, *PloS one*, 15(2), e0228862.
10. R. Das Gupta, **IH. Sajal**, M. Hasan, I. Sutradhar, MR. Haider, M. Sarker, (2019), *Frequency of television viewing and association with overweight and obesity among women of the reproductive age group in Myanmar: results from a nationwide cross-sectional survey*, *BMJ Open*, 9(3), e024680.

PROFESSIONAL AFFILIATION

Member - American Statistical Association	Jun. 2021 - present
Associate Member - American Association for Cancer Research	Jan. 2026 - present

MEDIA APPEARANCES

- www.keranews.org/health-wellness/2022-10-28/ut-dallas-racial-disparities-breast-cancer-survivor-risk-screening
- magazine.utdallas.edu/2022/10/20/utd-researchers-address-racial-disparity-in-breast-cancer-risk-assessment/
- nsm.utdallas.edu/biswas-tool-black-breast-cancer-survivors/